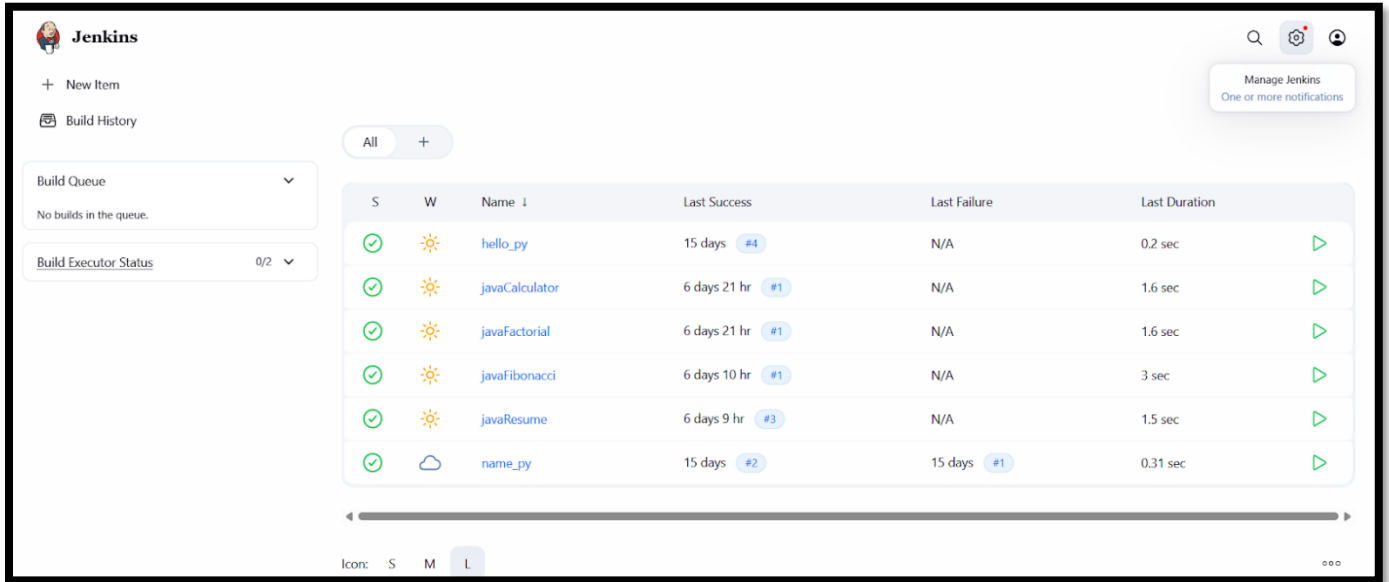


Experiment 6:

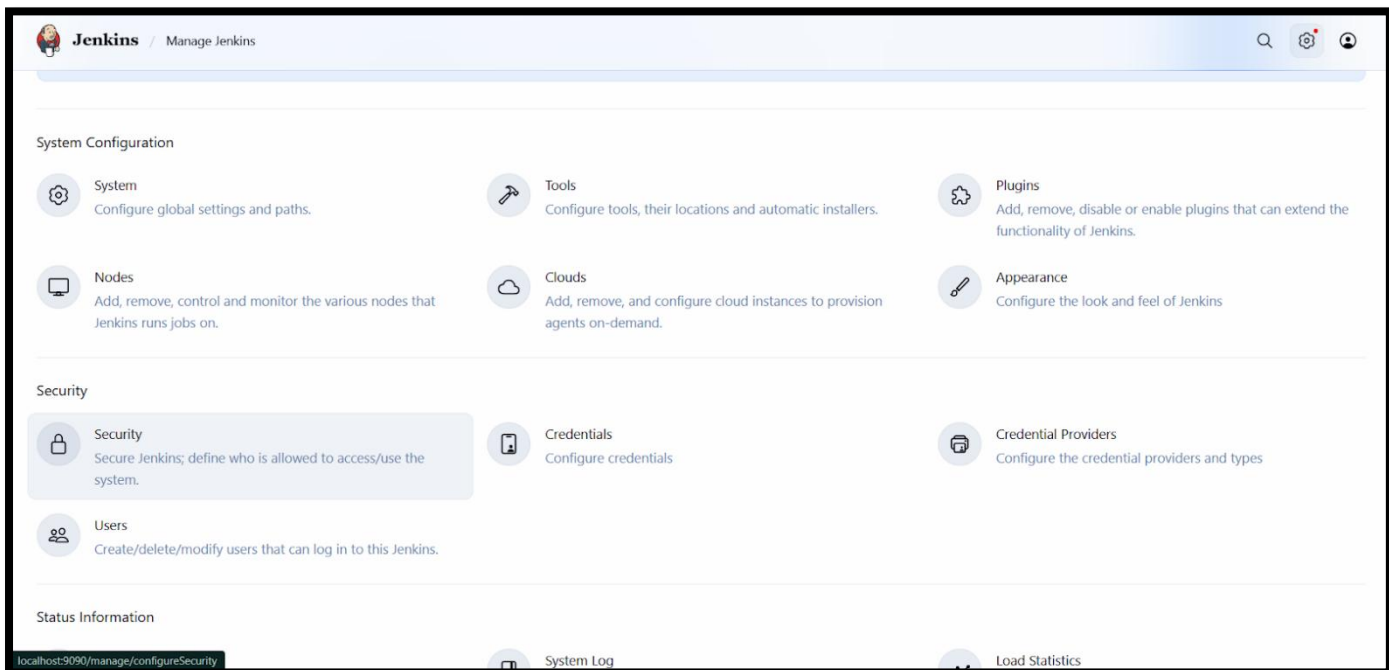
1. Accessing Jenkins Dashboard and Opening Manage Jenkins Settings



The screenshot shows the Jenkins Dashboard. On the left, there's a sidebar with 'New Item', 'Build History', 'Build Queue' (showing 'No builds in the queue.'), and 'Build Executor Status' (showing '0/2'). The main area displays a table of builds with columns: S (Status), W (Webhook), Name, Last Success, Last Failure, and Last Duration. The table lists six builds: 'hello_py', 'javaCalculator', 'javaFactorial', 'javaFibonacci', 'javaResume', and 'name_py'. Each build has a green checkmark in the 'S' column and a sun icon in the 'W' column. The 'Last Success' column shows the time since the last successful build, and the 'Last Failure' column shows the time since the last failure. The 'Last Duration' column shows the duration of the last build. At the bottom, there's a pagination bar with 'Icon: S M L' and a '...' button.

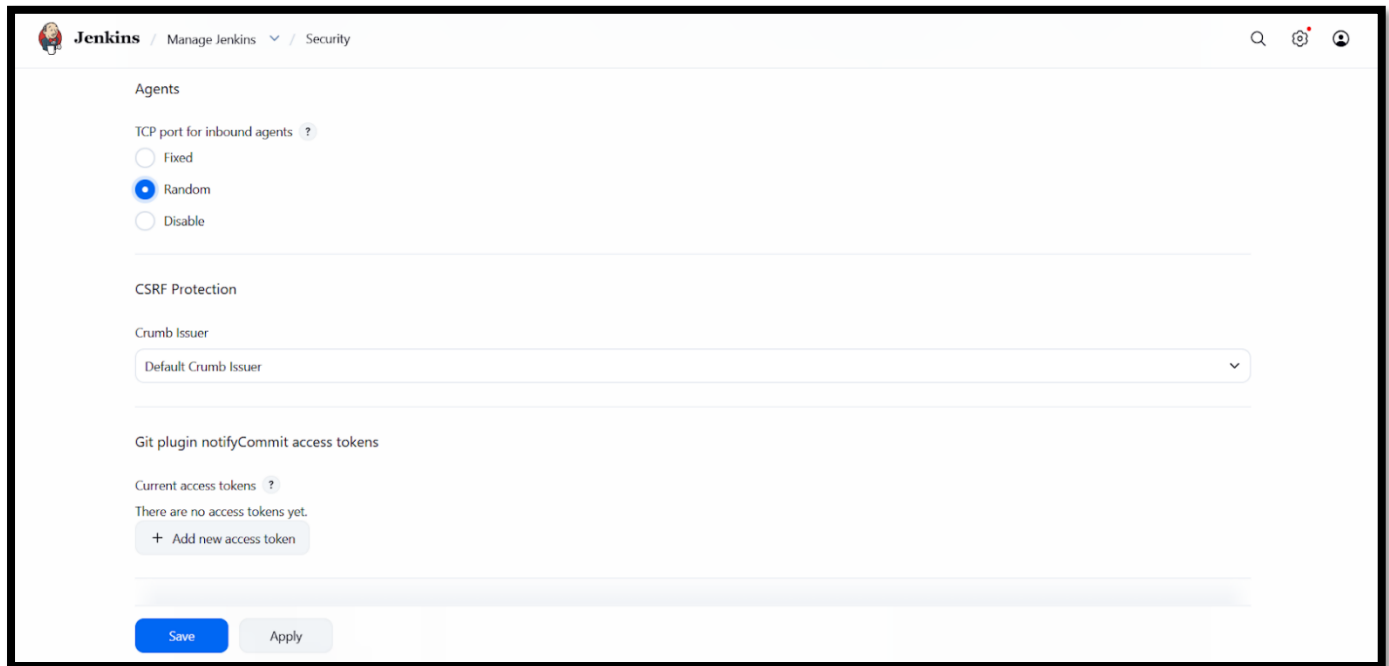
S	W	Name	Last Success	Last Failure	Last Duration
✓	☀	hello_py	15 days #4	N/A	0.2 sec
✓	☀	javaCalculator	6 days 21 hr #1	N/A	1.6 sec
✓	☀	javaFactorial	6 days 21 hr #1	N/A	1.6 sec
✓	☀	javaFibonacci	6 days 10 hr #1	N/A	3 sec
✓	☀	javaResume	6 days 9 hr #3	N/A	1.5 sec
✓	☁	name_py	15 days #2	15 days #1	0.31 sec

2. Configuring Jenkins Security Settings and TCP Port for Inbound Agents



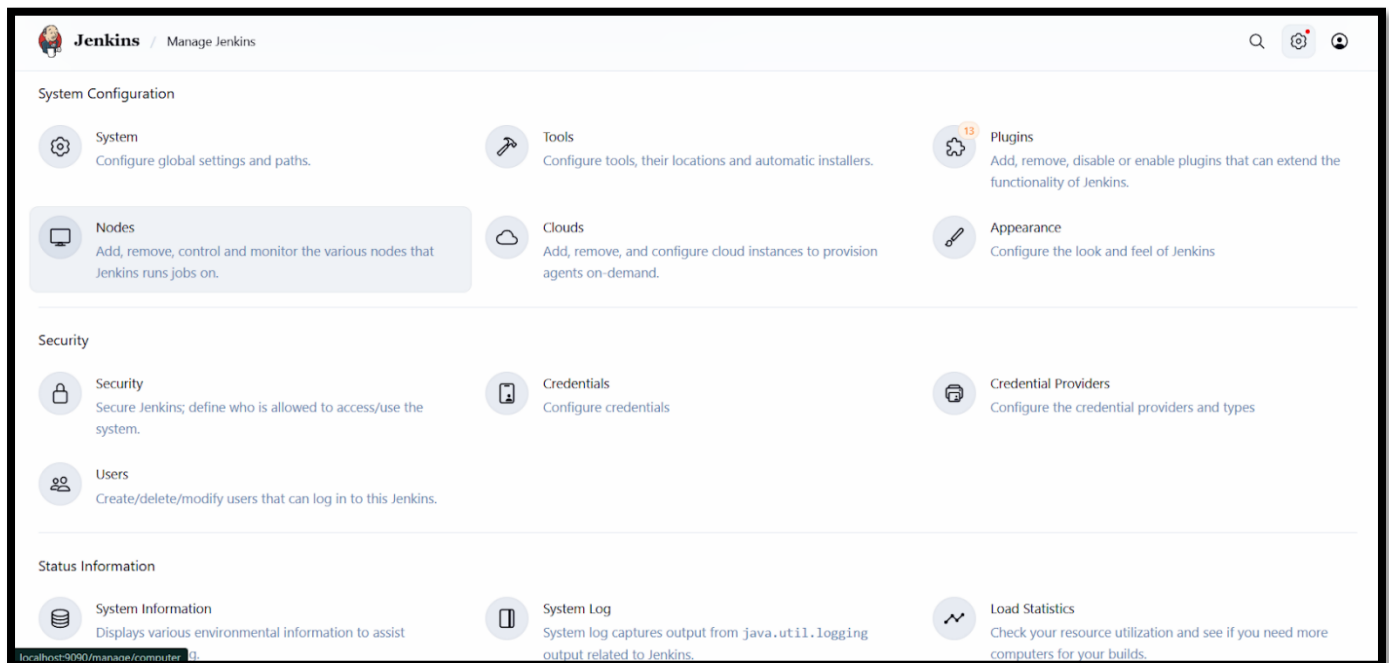
The screenshot shows the Jenkins 'Manage Jenkins' page. The top bar includes the Jenkins logo, 'Manage Jenkins', and search, settings, and user icons. The main content is divided into two sections: 'System Configuration' and 'Security'. The 'System Configuration' section includes 'System' (Configure global settings and paths), 'Tools' (Configure tools, their locations and automatic installers), 'Plugins' (Add, remove, disable or enable plugins that can extend the functionality of Jenkins), 'Nodes' (Add, remove, control and monitor the various nodes that Jenkins runs jobs on), 'Clouds' (Add, remove, and configure cloud instances to provision agents on-demand), and 'Appearance' (Configure the look and feel of Jenkins). The 'Security' section includes 'Security' (Secure Jenkins; define who is allowed to access/use the system), 'Credentials' (Configure credentials), 'Credential Providers' (Configure the credential providers and types), and 'Users' (Create/delete/modify users that can log in to this Jenkins). At the bottom, there's a 'Status Information' section with 'System Log' and 'Load Statistics'.

localhost:9090/manage/configureSecurity



The image shows the Jenkins 'Security' configuration page. It includes sections for 'Agents' (with radio buttons for 'Fixed', 'Random', and 'Disable'), 'CSRF Protection' (with a 'Crumb Issuer' dropdown set to 'Default Crumb Issuer'), and 'Git plugin notify/Commit access tokens' (with a '+ Add new access token' button). At the bottom are 'Save' and 'Apply' buttons.

3. Navigating to Jenkins Nodes Section to Manage Build Agents



The image shows the Jenkins 'Manage Jenkins' dashboard. It features several configuration categories: 'System Configuration' (System, Tools, Plugins, Appearance), 'Nodes' (Add, remove, control and monitor the various nodes that Jenkins runs jobs on), 'Security' (Security, Credentials, Credential Providers, Users), and 'Status Information' (System Information, System Log, Load Statistics).

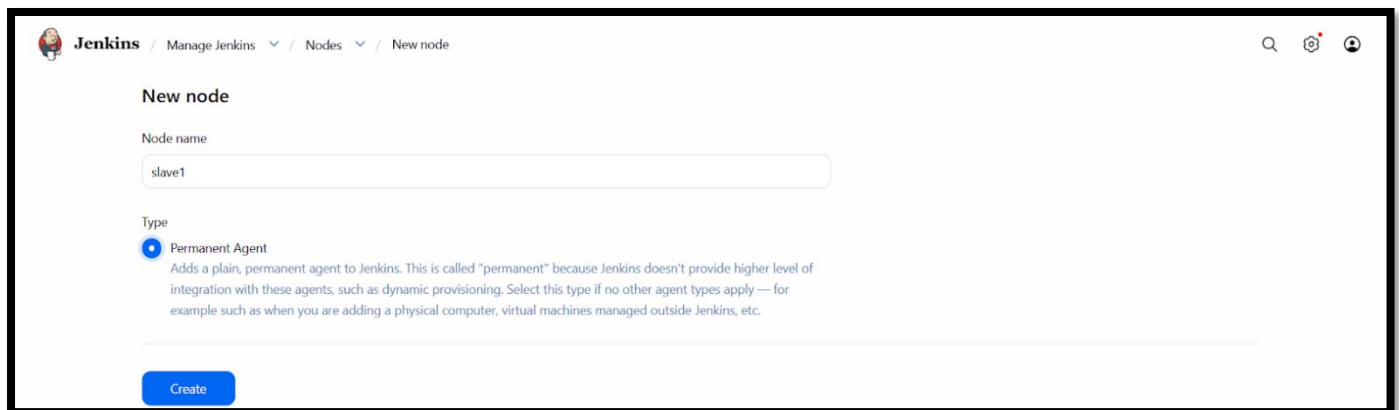


The image shows the Jenkins 'Nodes' page. It includes a '+ New Node' button and a 'Configure Monitors' button. Below is a table listing the current nodes.

S	Name	Architecture	Clock Difference	Free Disk Space	Free Swap Space	Free Temp Space	Response Time
	Built-In Node	Windows 11 (amd64)	In sync	314.34 GiB	10.47 GiB	314.34 GiB	0ms
	Data obtained	24 min	24 min	24 min	24 min	24 min	24 min

Icons: S M L Legend

4. Creating and Configuring a New Slave Node (Permanent Agent)



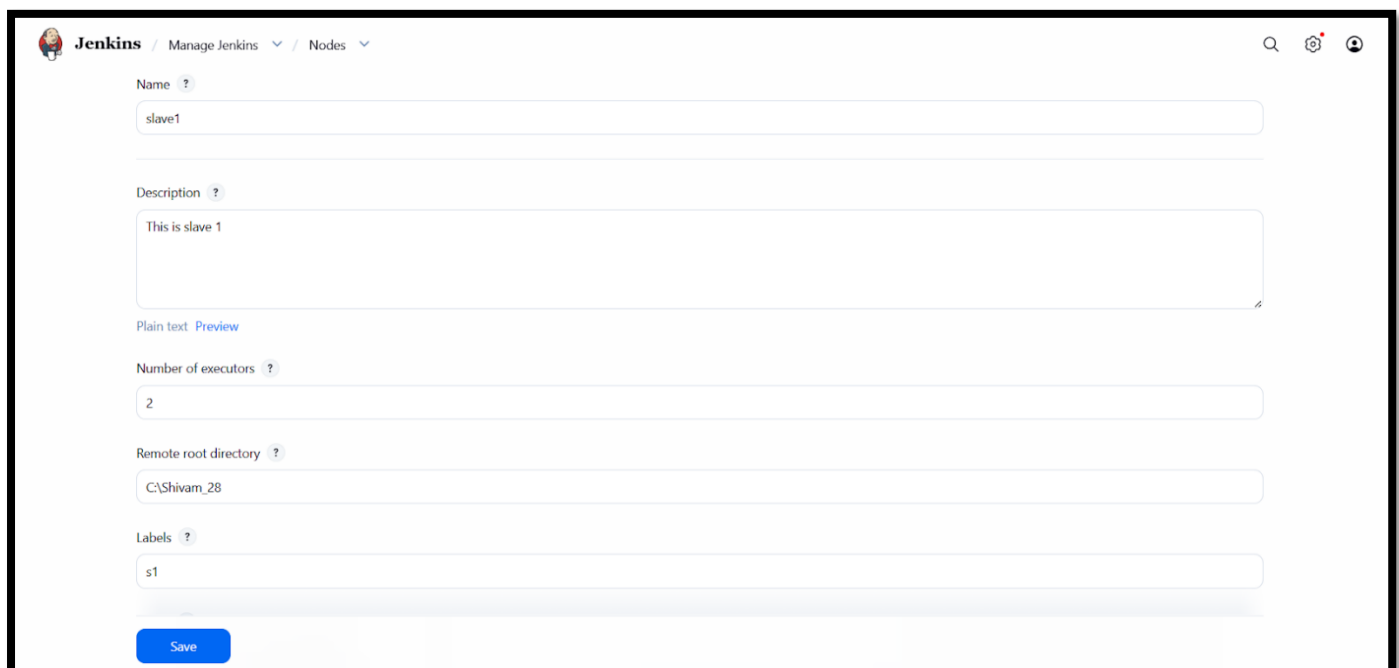
The screenshot shows the Jenkins 'New node' configuration page. The breadcrumb trail is 'Jenkins / Manage Jenkins / Nodes / New node'. The 'Node name' field contains 'slave1'. Under the 'Type' section, 'Permanent Agent' is selected with a radio button. A descriptive text explains that a permanent agent provides a higher level of integration with Jenkins, such as dynamic provisioning, and is selected when no other agent types apply, like for physical computers or virtual machines managed outside Jenkins. A blue 'Create' button is at the bottom.

New node

Node name
slave1

Type
☒ Permanent Agent
Adds a plain, permanent agent to Jenkins. This is called "permanent" because Jenkins doesn't provide higher level of integration with these agents, such as dynamic provisioning. Select this type if no other agent types apply — for example such as when you are adding a physical computer, virtual machines managed outside Jenkins, etc.

Create



The screenshot shows the Jenkins 'Node configuration' page for 'slave1'. The breadcrumb trail is 'Jenkins / Manage Jenkins / Nodes'. Fields include: 'Name' (slave1), 'Description' (This is slave 1), 'Plain text' (with a 'Preview' link), 'Number of executors' (2), 'Remote root directory' (C:\Shivam_28), and 'Labels' (s1). A blue 'Save' button is at the bottom.

Name ?
slave1

Description ?
This is slave 1

Plain text [Preview](#)

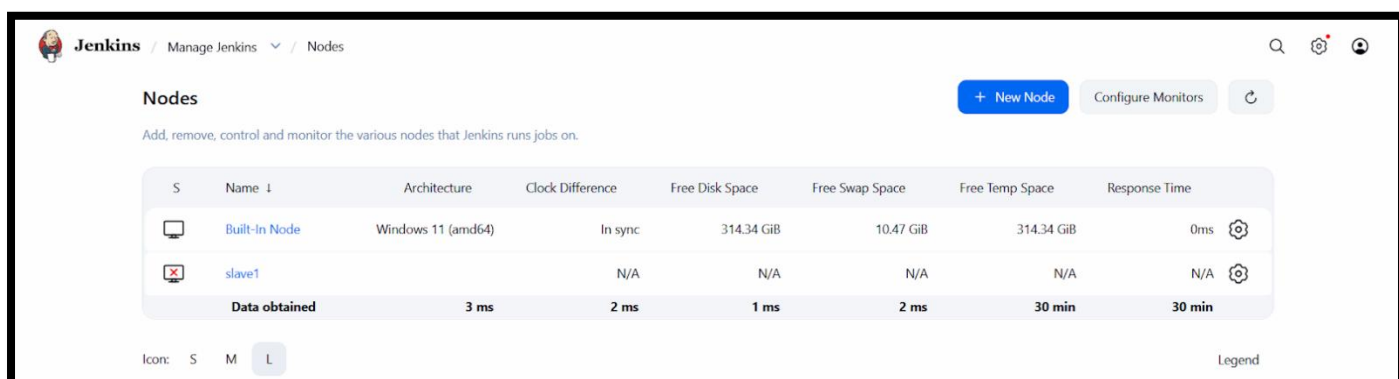
Number of executors ?
2

Remote root directory ?
C:\Shivam_28

Labels ?
s1

Save

5. Connecting Slave Node by Running Agent Command in CMD



The screenshot shows the Jenkins 'Nodes' page. It has a table with columns: S, Name, Architecture, Clock Difference, Free Disk Space, Free Swap Space, Free Temp Space, and Response Time. There are two nodes: 'Built-In Node' and 'slave1'. The 'slave1' node is highlighted with a red box. Below the table, there is a 'Data obtained' row showing values for each column. At the bottom, there is a legend for icons: S, M, and L.

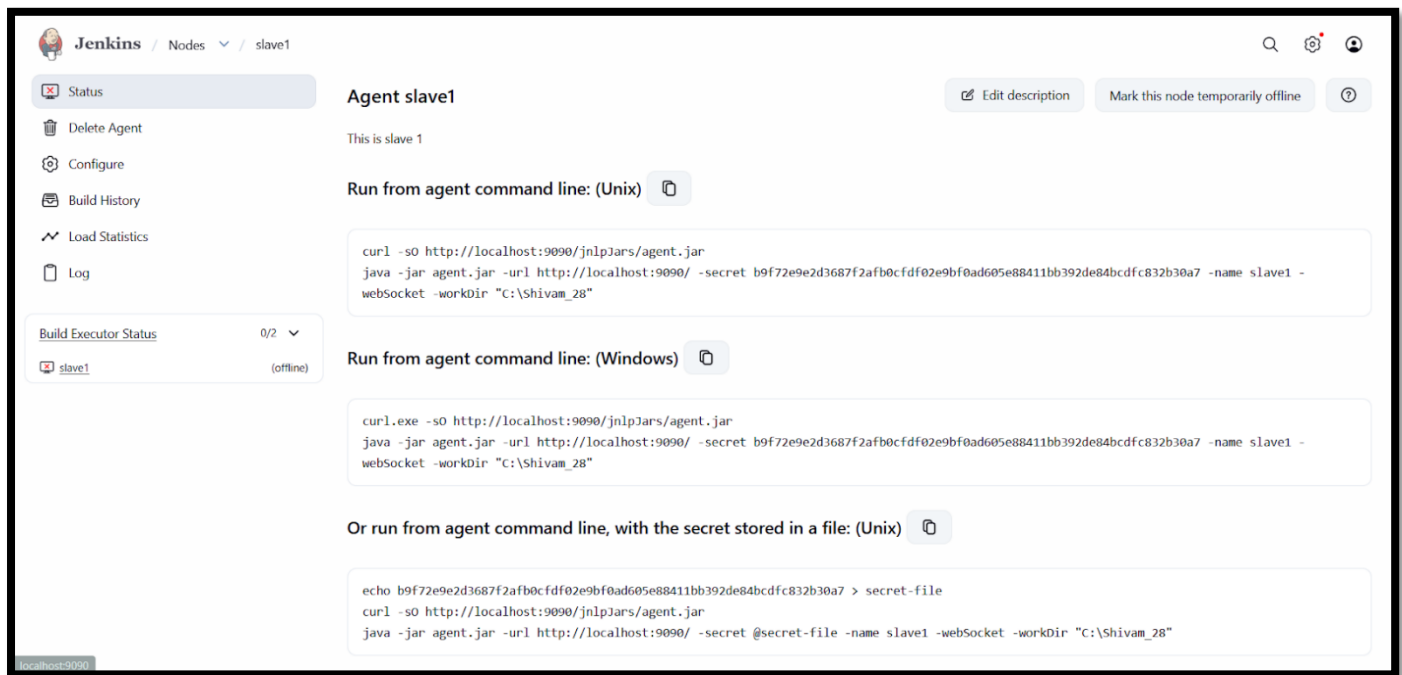
Nodes

Add, remove, control and monitor the various nodes that Jenkins runs jobs on.

S	Name	Architecture	Clock Difference	Free Disk Space	Free Swap Space	Free Temp Space	Response Time
	Built-In Node	Windows 11 (amd64)	In sync	314.34 GiB	10.47 GiB	314.34 GiB	0ms
	slave1		N/A	N/A	N/A	N/A	N/A
Data obtained		3 ms	2 ms	1 ms	2 ms	30 min	30 min

Icon: S M L

Legend



The screenshot shows the Jenkins 'Agent slave1' configuration page. The left sidebar contains links for Status, Delete Agent, Configure, Build History, Load Statistics, and Log. The main content area shows the agent's status as 'offline' and provides instructions for running the agent command line. It includes two code blocks: one for Unix and one for Windows, both showing the curl command to download the Jenkins agent jar and the java command to run it. A third code block shows an alternative method for Unix using a secret file.

Agent slave1

This is slave 1

Run from agent command line: (Unix)

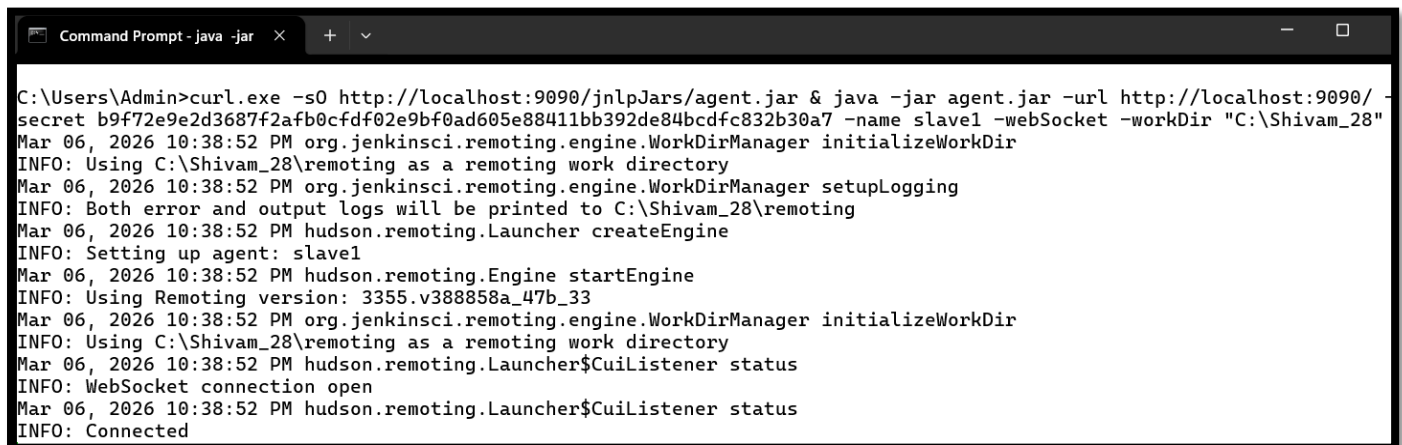
```
curl -sO http://localhost:9090/jnlpJars/agent.jar
java -jar agent.jar -url http://localhost:9090/ -secret b9f72e9e2d3687f2afb0cfd02e9bf0ad605e88411bb392de84bcdfc832b30a7 -name slave1 -webSocket -workDir "C:\Shivam_28"
```

Run from agent command line: (Windows)

```
curl.exe -sO http://localhost:9090/jnlpJars/agent.jar
java -jar agent.jar -url http://localhost:9090/ -secret b9f72e9e2d3687f2afb0cfd02e9bf0ad605e88411bb392de84bcdfc832b30a7 -name slave1 -webSocket -workDir "C:\Shivam_28"
```

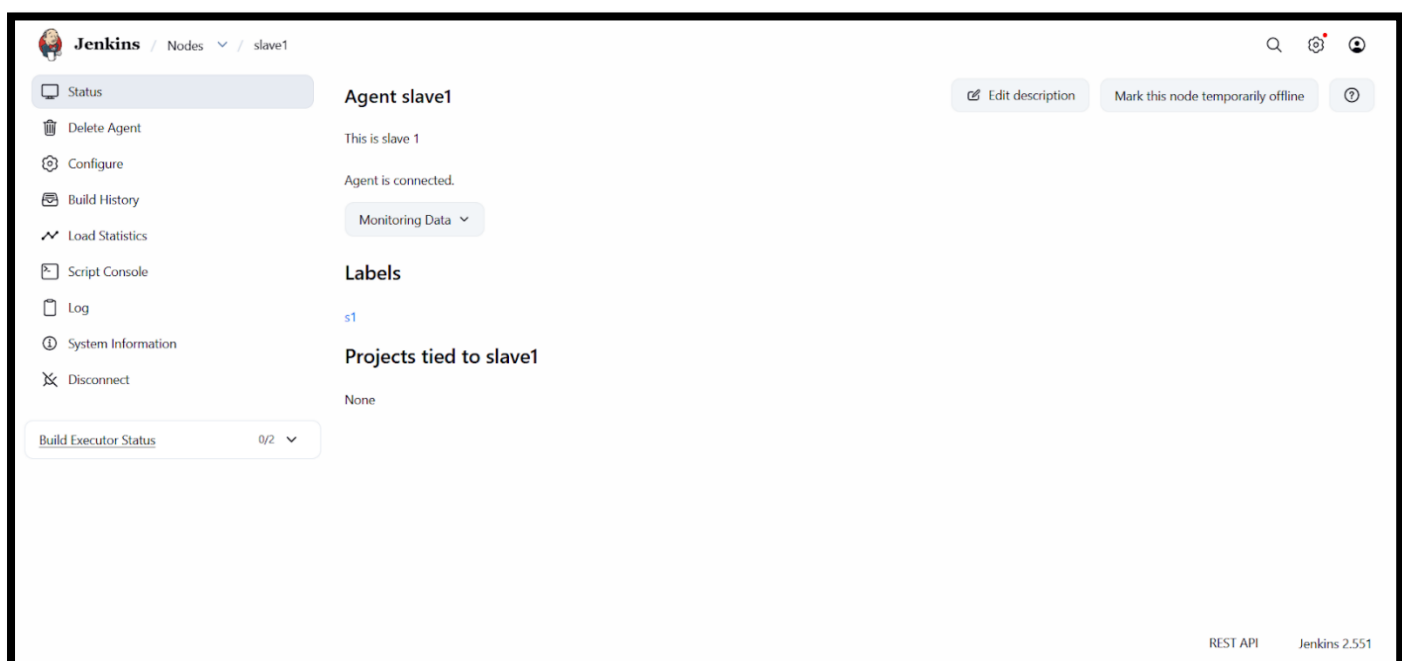
Or run from agent command line, with the secret stored in a file: (Unix)

```
echo b9f72e9e2d3687f2afb0cfd02e9bf0ad605e88411bb392de84bcdfc832b30a7 > secret-file
curl -sO http://localhost:9090/jnlpJars/agent.jar
java -jar agent.jar -url http://localhost:9090/ -secret @secret-file -name slave1 -webSocket -workDir "C:\Shivam_28"
```



The screenshot shows a Windows Command Prompt window with the following output:

```
C:\Users\Admin>curl.exe -sO http://localhost:9090/jnlpJars/agent.jar & java -jar agent.jar -url http://localhost:9090/ -secret b9f72e9e2d3687f2afb0cfd02e9bf0ad605e88411bb392de84bcdfc832b30a7 -name slave1 -webSocket -workDir "C:\Shivam_28"
Mar 06, 2026 10:38:52 PM org.jenkinsci.remoting.engine.WorkDirManager initializeWorkDir
INFO: Using C:\Shivam_28\remoting as a remoting work directory
Mar 06, 2026 10:38:52 PM org.jenkinsci.remoting.engine.WorkDirManager setupLogging
INFO: Both error and output logs will be printed to C:\Shivam_28\remoting
Mar 06, 2026 10:38:52 PM hudson.remoting.Launcher createEngine
INFO: Setting up agent: slave1
Mar 06, 2026 10:38:52 PM hudson.remoting.Engine startEngine
INFO: Using Remoting version: 3355.v388858a_47b_33
Mar 06, 2026 10:38:52 PM org.jenkinsci.remoting.engine.WorkDirManager initializeWorkDir
INFO: Using C:\Shivam_28\remoting as a remoting work directory
Mar 06, 2026 10:38:52 PM hudson.remoting.Launcher$CuiListener status
INFO: WebSocket connection open
Mar 06, 2026 10:38:52 PM hudson.remoting.Launcher$CuiListener status
INFO: Connected
```



The screenshot shows the Jenkins 'Agent slave1' configuration page after the agent has been successfully connected. The status is now 'connected'. The 'Monitoring Data' dropdown is expanded, showing 's1' as the label. The 'Projects tied to slave1' section is empty.

Agent slave1

This is slave 1

Agent is connected.

Monitoring Data

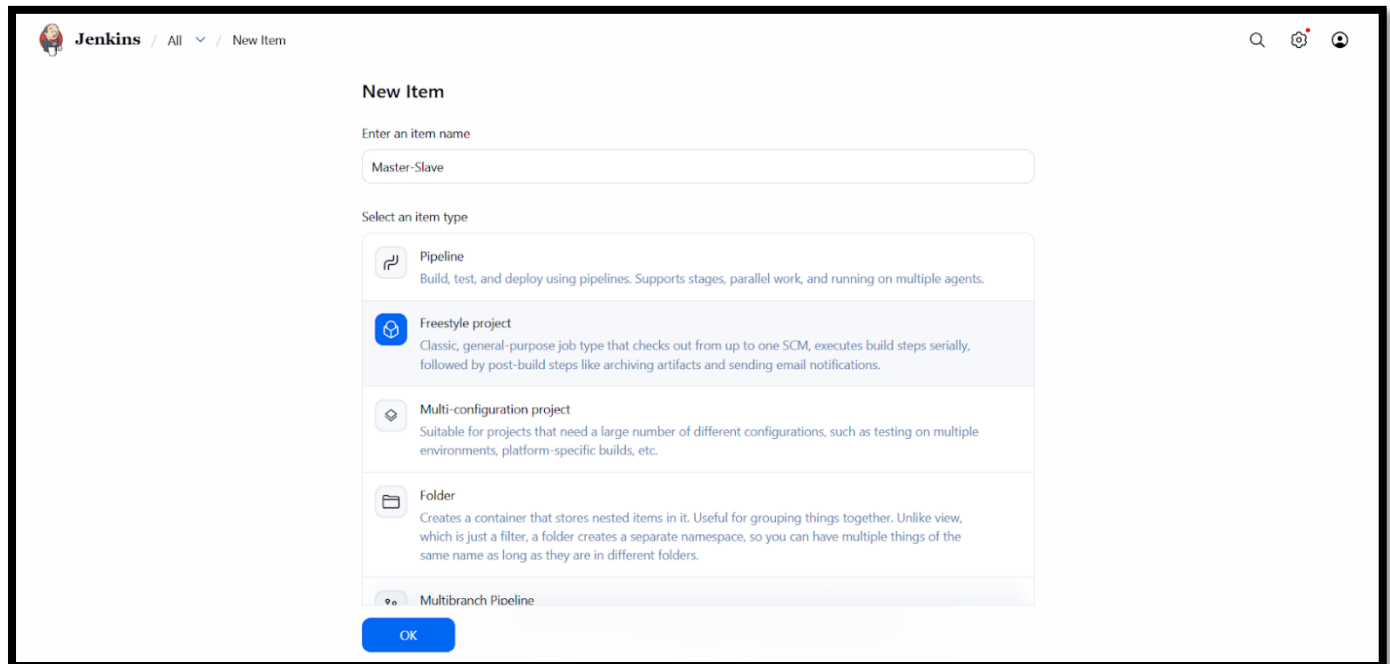
Labels

s1

Projects tied to slave1

None

6. Creating and Configuring a Freestyle Project for Master–Slave Build Execution



The screenshot shows the Jenkins 'New Item' page. At the top, the breadcrumb is 'Jenkins / All / New Item'. The 'Enter an item name' field contains 'Master-Slave'. Under 'Select an item type', four options are listed: Pipeline, Freestyle project (selected), Multi-configuration project, and Folder. A 'Multibranch Pipeline' option is partially visible at the bottom. An 'OK' button is at the bottom right.

New Item

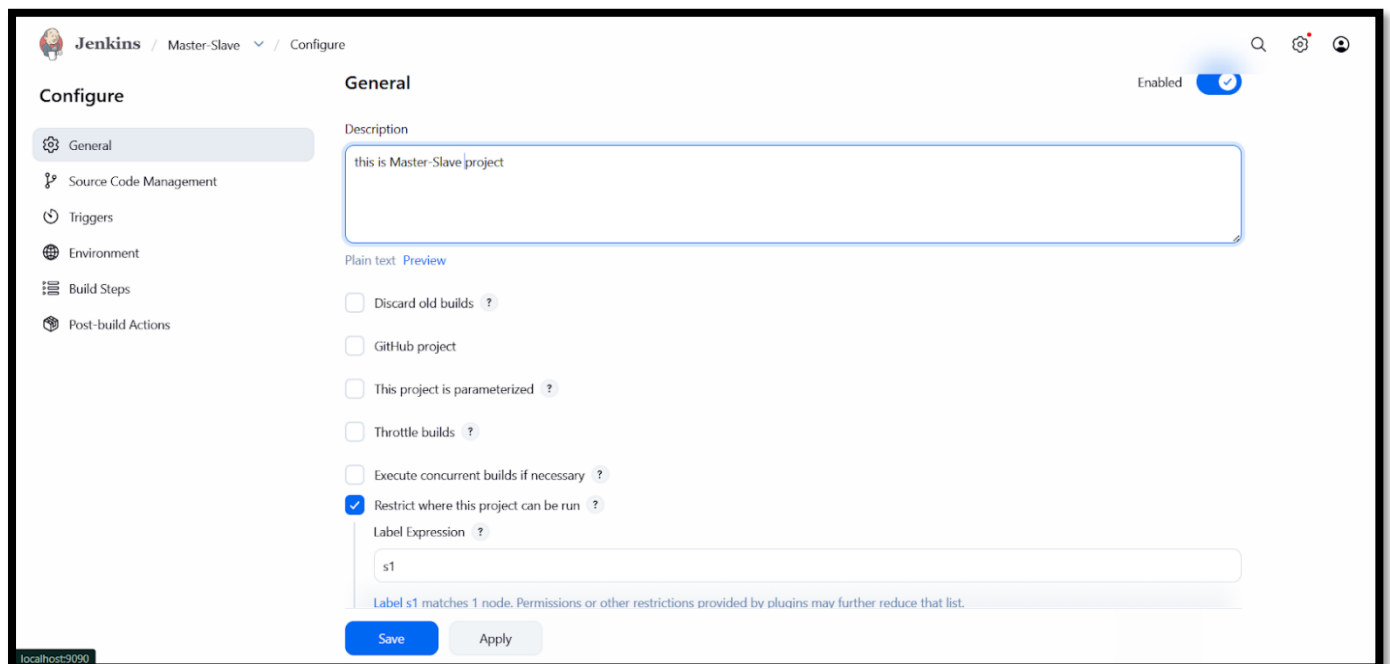
Enter an item name

Master-Slave

Select an item type

- Pipeline**
Build, test, and deploy using pipelines. Supports stages, parallel work, and running on multiple agents.
- Freestyle project**
Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.
- Multi-configuration project**
Suitable for projects that need a large number of different configurations, such as testing on multiple environments, platform-specific builds, etc.
- Folder**
Creates a container that stores nested items in it. Useful for grouping things together. Unlike view, which is just a filter, a folder creates a separate namespace, so you can have multiple things of the same name as long as they are in different folders.
- Multibranch Pipeline**

OK



The screenshot shows the Jenkins configuration page for the 'Master-Slave' project. The breadcrumb is 'Jenkins / Master-Slave / Configure'. The 'General' tab is selected in the left sidebar. The 'Description' field contains 'this is Master-Slave project'. The 'Label Expression' field contains 's1'. The 'Restrict where this project can be run' checkbox is checked. The 'Enabled' toggle is turned on. 'Save' and 'Apply' buttons are at the bottom.

Configure

General

Source Code Management

Triggers

Environment

Build Steps

Post-build Actions

General

Enabled

Description

this is Master-Slave project

Plain text [Preview](#)

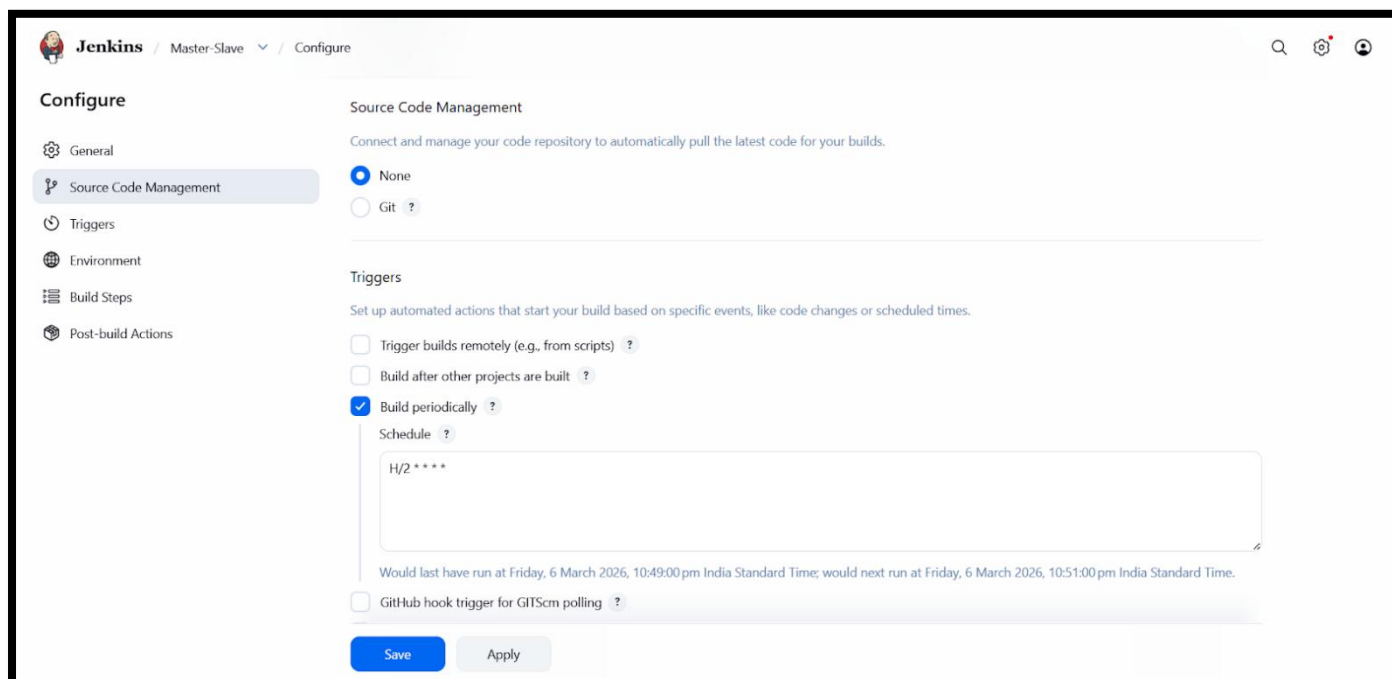
- ☐ Discard old builds ?
- ☐ GitHub project
- ☐ This project is parameterized ?
- ☐ Throttle builds ?
- ☐ Execute concurrent builds if necessary ?
- ☒ Restrict where this project can be run ?

Label Expression ?

s1

Label s1 matches 1 node. Permissions or other restrictions provided by plugins may further reduce that list.

Save Apply



The image shows the Jenkins configuration page for a job named 'Master-Slave'. The 'Configure' tab is selected, and the 'Source Code Management' section is active. Under 'Source Code Management', the 'None' option is selected. The 'Triggers' section is also active, showing options to trigger builds remotely, build after other projects, or build periodically. The 'Build periodically' option is checked, and the schedule is set to 'H/2 * * * *'. The page includes a 'Save' button and an 'Apply' button.

Jenkins / Master-Slave / Configure

Configure

- General
- Source Code Management**
- Triggers
- Environment
- Build Steps
- Post-build Actions

Source Code Management

Connect and manage your code repository to automatically pull the latest code for your builds.

☒ None
☐ Git ?

Triggers

Set up automated actions that start your build based on specific events, like code changes or scheduled times.

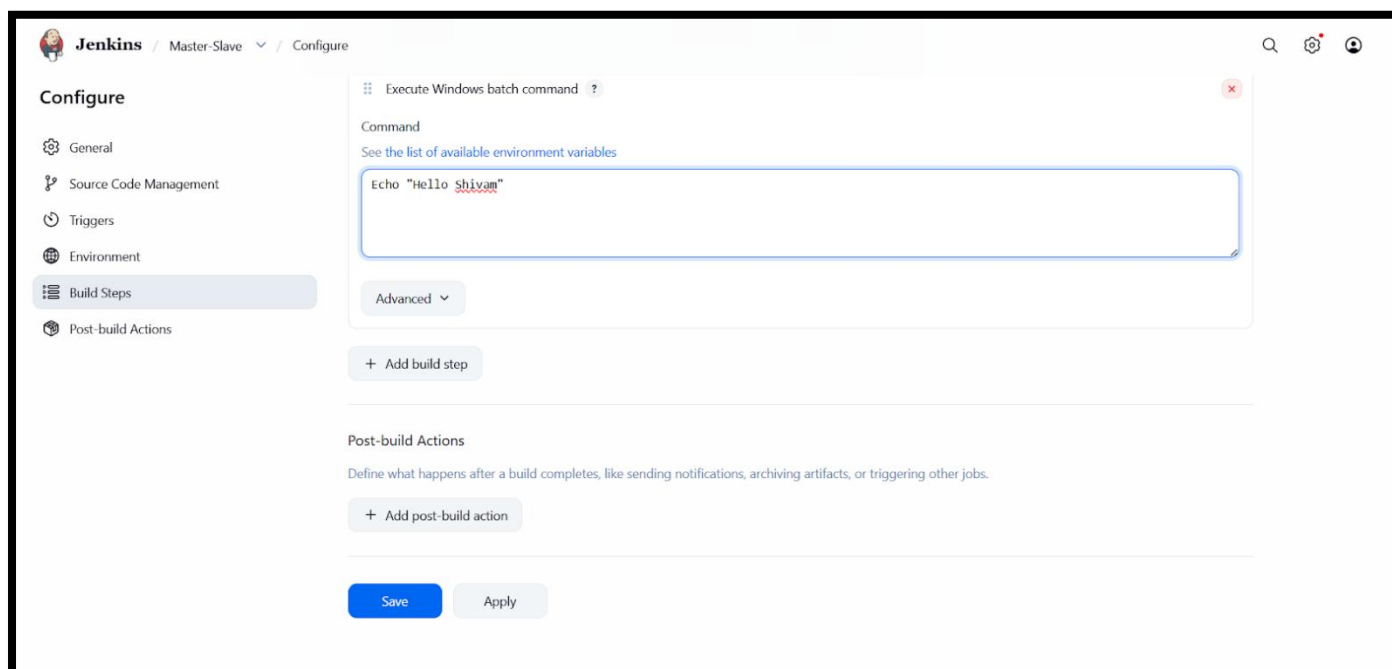
☐ Trigger builds remotely (e.g., from scripts) ?
☐ Build after other projects are built ?
☒ Build periodically ?

Schedule ?
H/2 * * * *

Would last have run at Friday, 6 March 2026, 10:49:00 pm India Standard Time; would next run at Friday, 6 March 2026, 10:51:00 pm India Standard Time.

☐ GitHub hook trigger for GITScm polling ?

Save **Apply**



The image shows the Jenkins configuration page for a job named 'Master-Slave'. The 'Configure' tab is selected, and the 'Build Steps' section is active. A new build step is added, titled 'Execute Windows batch command'. The command is set to 'Echo "Hello Shivam"'. The 'Post-build Actions' section is also visible, showing an option to add post-build actions. The page includes a 'Save' button and an 'Apply' button.

Jenkins / Master-Slave / Configure

Configure

- General
- Source Code Management
- Triggers
- Environment
- Build Steps**
- Post-build Actions

Execute Windows batch command ?

Command
See the list of available environment variables

Echo "Hello Shivam"

Advanced ▾

+ Add build step

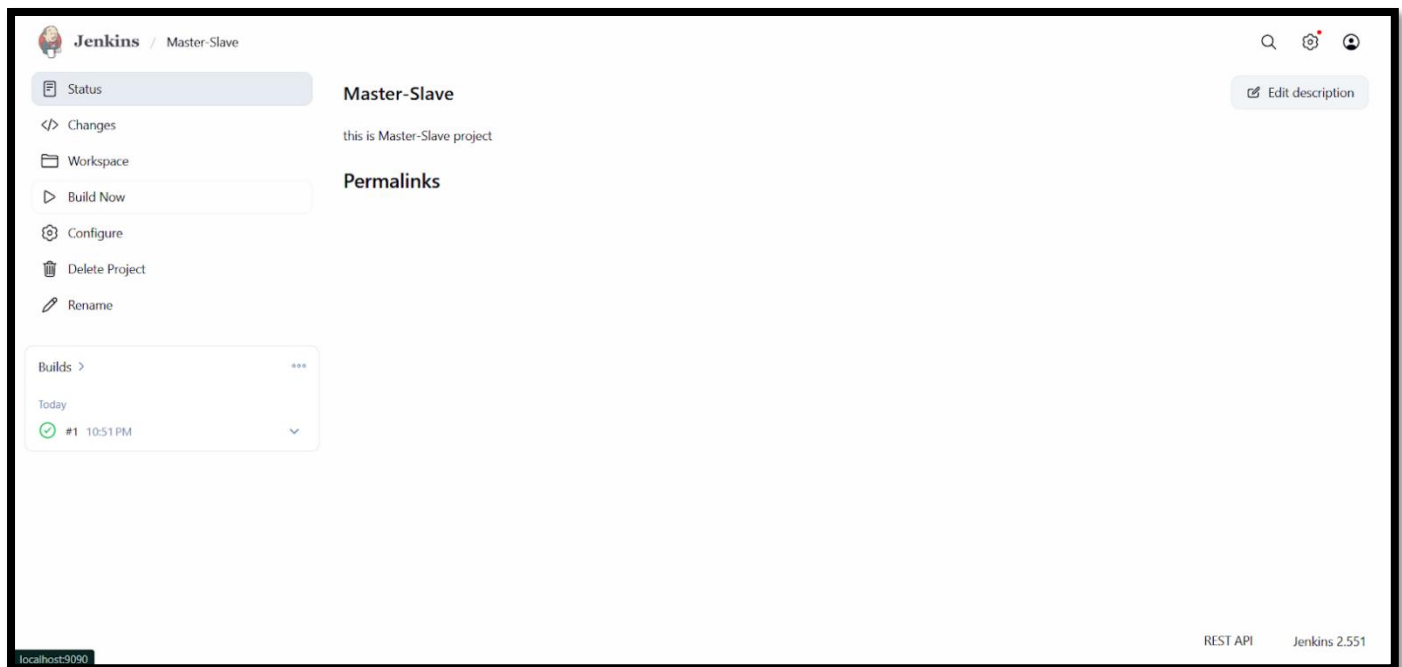
Post-build Actions

Define what happens after a build completes, like sending notifications, archiving artifacts, or triggering other jobs.

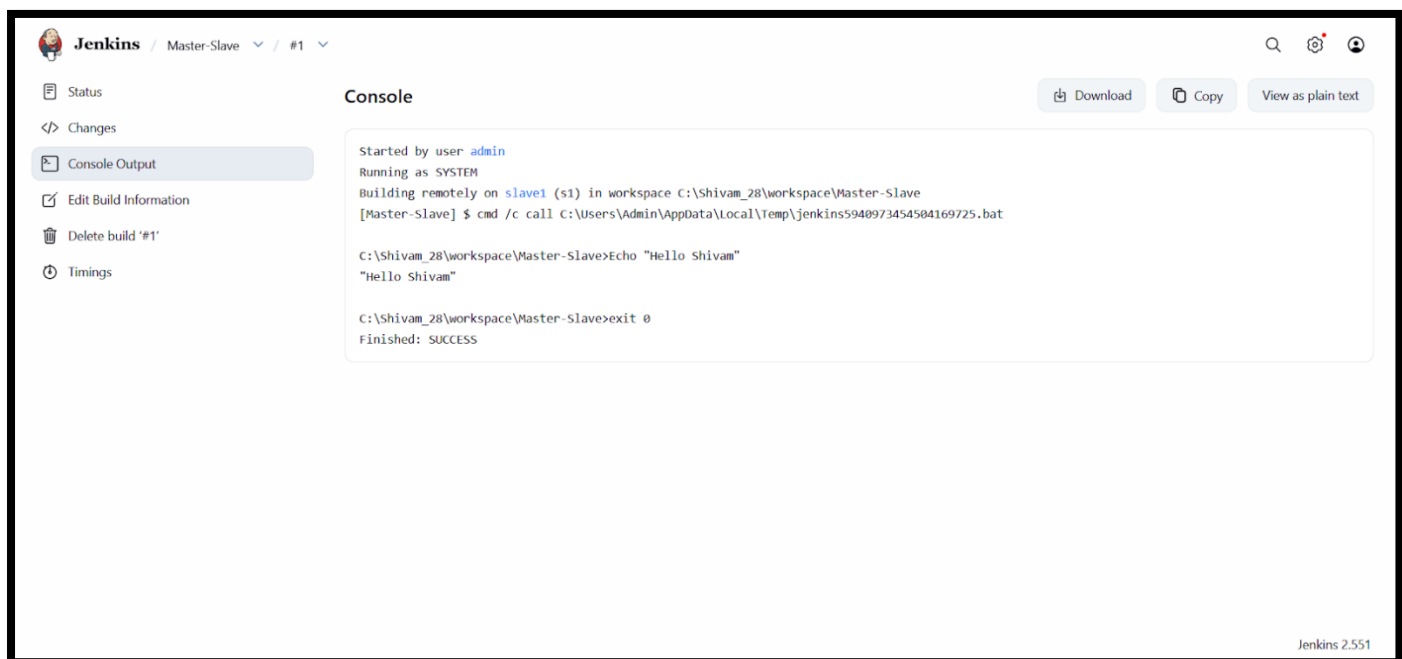
+ Add post-build action

Save **Apply**

7. Executing Master-Slave Job and Viewing Successful Build Output



The screenshot shows the Jenkins web interface for a job named 'Master-Slave'. The left sidebar contains navigation links: Status, Changes, Workspace, Build Now, Configure, Delete Project, and Rename. The main content area displays the job's status as 'this is Master-Slave project' and a 'Permalinks' section. Below this, a 'Builds' section shows a single build (#1) completed at 10:51 PM. The bottom status bar indicates 'localhost:9090', 'REST API', and 'Jenkins 2.551'.



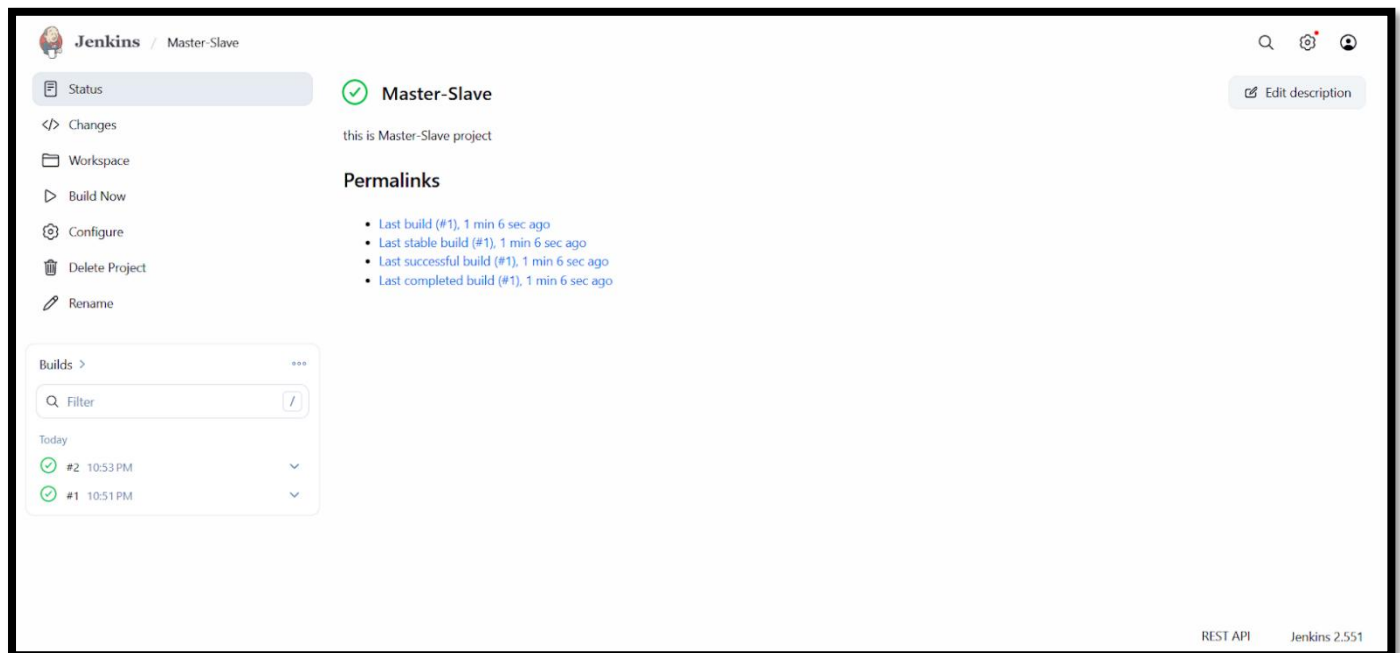
The screenshot shows the Jenkins web interface for the 'Master-Slave' job, specifically the console output for build #1. The left sidebar includes links for Status, Changes, Console Output, Edit Build Information, Delete build '#1', and Timings. The main content area displays the console output, which includes the following text:

```
Started by user admin
Running as SYSTEM
Building remotely on slave1 (s1) in workspace C:\Shivam_28\workspace\Master-Slave
[Master-Slave] $ cmd /c call C:\Users\Admin\AppData\Local\Temp\jenkins5940973454504169725.bat

C:\Shivam_28\workspace\Master-Slave>Echo "Hello Shivam"
"Hello Shivam"

C:\Shivam_28\workspace\Master-Slave>exit 0
Finished: SUCCESS
```

The bottom status bar indicates 'Jenkins 2.551'.



The screenshot shows the Jenkins web interface for a project named 'Master-Slave'. The left sidebar contains navigation links: Status (selected), Changes, Workspace, Build Now, Configure, Delete Project, and Rename. The main content area features a green checkmark icon and the title 'Master-Slave', followed by the text 'this is Master-Slave project'. Below this is a 'Permalinks' section with four links: 'Last build (#1), 1 min 6 sec ago', 'Last stable build (#1), 1 min 6 sec ago', 'Last successful build (#1), 1 min 6 sec ago', and 'Last completed build (#1), 1 min 6 sec ago'. A 'Builds' section on the left includes a search filter and a list of two builds: '#2 10:53 PM' and '#1 10:51 PM', both marked with green checkmarks. The bottom right corner displays 'REST API' and 'Jenkins 2.551'.

Jenkins / Master-Slave

Status

Master-Slave

this is Master-Slave project

Permalinks

- Last build (#1), 1 min 6 sec ago
- Last stable build (#1), 1 min 6 sec ago
- Last successful build (#1), 1 min 6 sec ago
- Last completed build (#1), 1 min 6 sec ago

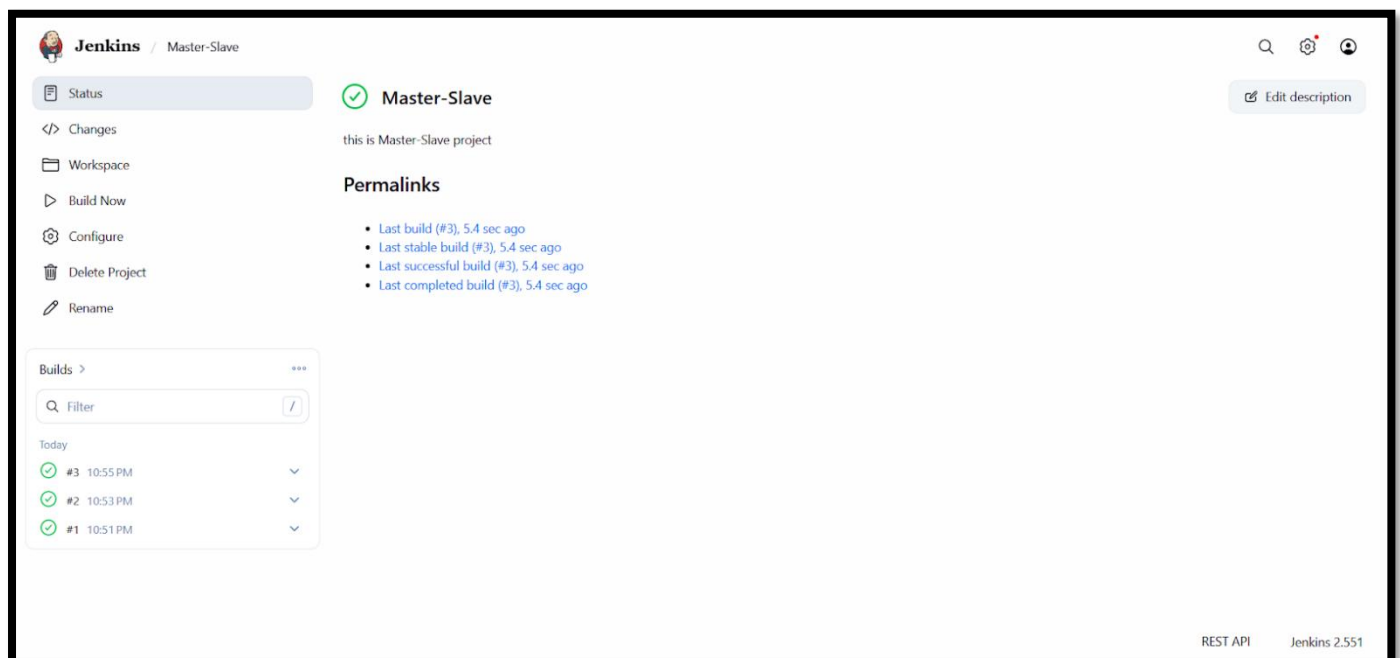
Builds >

Filter

Today

- #2 10:53 PM
- #1 10:51 PM

REST API Jenkins 2.551



This screenshot is similar to the one above but shows three builds instead of two. The 'Builds' section now lists '#3 10:55 PM', '#2 10:53 PM', and '#1 10:51 PM', all with green checkmarks. The 'Permalinks' section has been updated to reflect the third build, with links for '#3' instead of '#1'. The rest of the interface, including the sidebar and footer, remains the same.

Jenkins / Master-Slave

Status

Master-Slave

this is Master-Slave project

Permalinks

- Last build (#3), 5.4 sec ago
- Last stable build (#3), 5.4 sec ago
- Last successful build (#3), 5.4 sec ago
- Last completed build (#3), 5.4 sec ago

Builds >

Filter

Today

- #3 10:55 PM
- #2 10:53 PM
- #1 10:51 PM

REST API Jenkins 2.551